

TS154G – TS155G

Falcon Telemetry

General Manual



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1 INTRODUCTION

TS154G is a 4G communication device designed for real-time transmission of telemetry data. Through a simple Internet connection, the device connects to the Power On servers, allowing the reception and display of data from its own telemetry; data transmission can take place from any location, without geographical constraints.

TS155G further improves performance thanks to the 5G connection and the presence of two modems working in parallel, providing strength and reliability even in the most extreme conditions.

Thanks to this data transmission technology, the use of radio transmitters or other additional systems is no longer necessary, simplifying installation and reducing the complexity of the system.

The system is also compatible with any device equipped with an RS232 or Ethernet communication interface, ensuring high integration flexibility.

2 REQUESTED HARDWARE

Power On telemetry operational Kit:

- TS154G/TS155G telemetry modem
- SIM card with internet access
- Electrical wiring interface with the vehicle
- Antenna/s

Inside the box a stable Internet connection is required for normal telemetry use.

During the configuration phase, a commercial Ethernet cable is required to be connected between the PC and the telemetry modem.

3 SETUP PROCEDURE

Below are the steps required to make the telemetry system operational. For details on the various points, follow the link to the specific chapter.:

- 1) Verify that the SIM card you purchased is enabled for internet traffic and is not PIN-locked. To verify this, insert it into a mobile phone and remove the PIN requirement (turn the phone off and on again to verify). If necessary, enable roaming traffic.
- 2) Configure the car logger to enable telemetry data transmission (RS232 o ETH)
- 3) Install the PowerOn Bridge software on your PC (3.1.0.0 or superior, available on <https://t1.powerontelemetry.it/>)
- 4) Run PowerOn Bridge e configure it (see [PowerOnBridge configuration](#))
- 5) After inserting the SIM and turning on the telemetry module, connect to the module and proceed with the APN configuration based on the service provider (see [Car configuration](#))
- 6) If the configuration was successful, both the fixed blue PWR LED (indicating that the module is receiving power correctly) and the flashing green SYS LED (indicating that the module is transmitting data) should be lit on the telemetry module).
- 7) If everything has been done correctly, it is possible to see the data coming from the car in real time.

4 PIN OUT FOR TS154G AND TS155G

CONNECTOR	
Device	AS010-35PN
Mating (loom)	AS610-35SN
PIN OUT	
1	+12V
2	GND
3	RS232 RX (telemetry side)
4	N. C.
5	ETH1 TX+
6	ETH1 TX-
7	ETH1 RX+
8	ETH1 RX-
9	ETH2 TX+
10	ETH2 TX-
11	ETH2 RX+
12	ETH2 RX-
13	N.C.

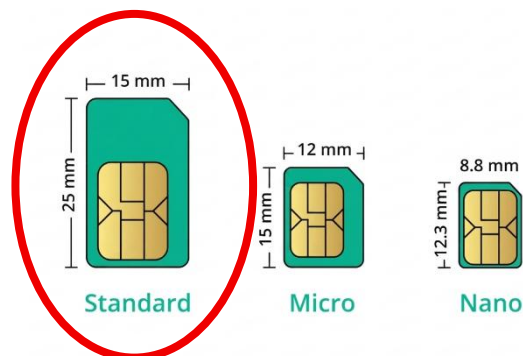
5 SIM CARD

You can use a SIM card from any mobile operator with an Internet connection.

You can also use the internet while roaming if your Internet contract allows it.

Make sure you have disabled your SIM security PIN.

A standard SIM is needed (2FF), for smaller formats use an adapter.



6 ANTENNAS

The TS154G telemetry module has two SMA connectors for the antennas, Primary (**PRI**) and Diversity (**DIV**).

For the correct functioning of the module, it is mandatory to connect an antenna to the **PRI** connector.

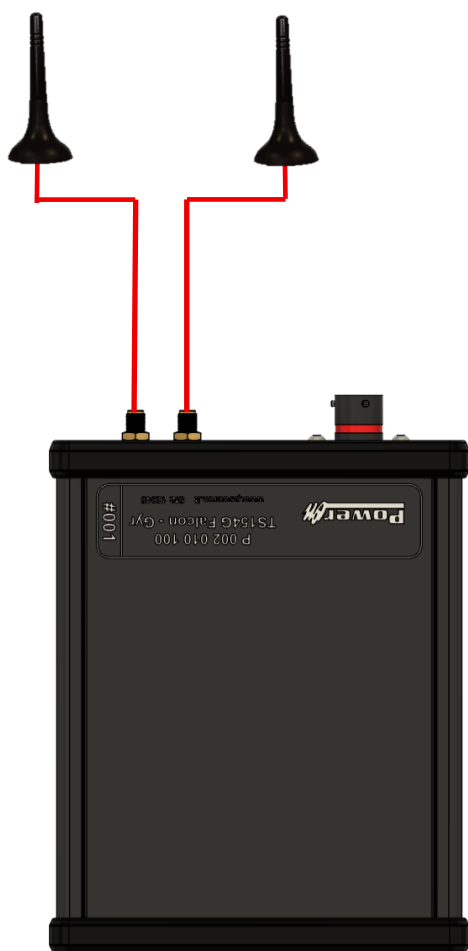
Connecting a second antenna to the **Diversity** connector is optional although highly recommended: its use allows to improve the reception quality, since the signal coming from the second antenna is combined with the main one, helping to reduce interference and attenuation.



The TS155G version, by doubling the number of modems, doubles the number of antennas while maintaining the same operating logic as the TS154G.

The diagram below summarizes how to make antenna connections on both types of telemetry (TS154G e TS155G).





TS154G - Gyr



TS155G - Lanner

The antennas must be positioned away from other antennas on the car (GPS, Radio, etc.); furthermore, if you have two telemetry antennas, position them at least 20 cm apart.

7 STATUS LEDs

The telemetry status LEDs are located on the opposite side of the connectors. For around 20 second after turning on the device, both the LEDs will be on, then the behaviour of these LEDs reflects the status of the device as follows:

- LED PWR (blue): when it is on, the telemetry module is powered.
- LED SYS (green)
 - Slow flashing (4 Hz): connection initialization.
 - Fast flashing (10 Hz): connection established.



7.1 Data transmission speed and IP configuration

Below is a summary table of the main manufacturers of Motorsport acquisition systems and their respective data transmission speeds:

DEVICE	RS232 SUPPORT	ETH SUPPORT
Bosch	Max 115.2 kbaud/s	Max 200kb/s
Cosworth	Max 96 kbaud/s	Max 112kb/s
Magneti Marelli	Max 115.2 kbaud/s	-
MoTeC	Max 115.2 kbaud/s	Not available

Different devices need different IP address for the ethernet data sending; If you have a **Bosch logger**, configure it to send telemetry data to **IP 10.10.0.25, port 10001**. Otherwise, with a **Cosworth** logger, configure it to send the telemetry data to **IP 172.16.46.200, port 10001**.

If you have the second ethernet channel, keep the same IP, Bosch or Cosworth, as the first ethernet channel, just set the second port to **10002**.

8 CAR CONFIGURATION

8.1 Hardware

1. RJ45 Ethernet cable male-male



8.2 APN setup procedure

1. Telemetry must already be turned on and have completed the initial boot phase.
2. Connect the PC to the telemetry Ethernet connector, located on the back next to the LEDs. Remove the cap to access the connector.
3. On your PC, open a browser and connect to the IP address **192.168.1.100**. The following window will appear:

Authorization Required

Username

Password

4. Login:
Username: TS15
Password: ts15
It is possible to change the password after the first login on the **Telemetry -> Configuration** tab.

5. In the "Telemetry/Configuration" tab you can enter the desired APN in the Sim Card section.
6. After any change you do on the telemetry settings, press "Save" to save the changes

PowerOnTS15
Status
Telemetry
Diagnostic
Reboot
Log out

Telemetry configuration

When the telemetry is connected PowerOn will collect signal quality and modem data in order to improve the service. Please refer to our customer support for further informations

Login password

Telemetry password

Telemetry login password (default ts15), reboot required.

SIM card configuration

SIM card APN #1

APN to connect the first telemetry modem to internet. Use apn "po" if the SIM is not present

Telemetry license

Write the license key in the box corresponding to the streaming channel you want to enable. THE RS232 CHANNEL IS ALWAYS ENABLED. To enable Ethernet channels please contact PowerOn.

Telemetry enabled modules (other than RS232) **ETHERNET**

RS232 main license

RS232 license. Configure your logger to send data to RS232 with baudrate 115200

Ethernet 1 license

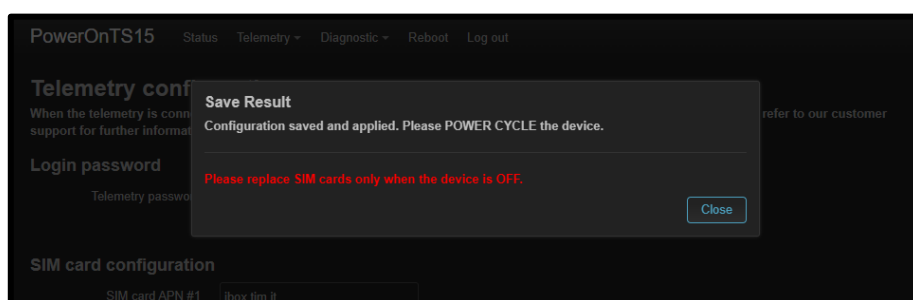
Ethernet license for channel 1. Configure your BOSCH logger to send telemetry data to IP 10.10.0.25 port 10001 or your COSWORTH logger to send telemetry data to IP 172.16.46.200 port 10001

Ethernet 2 license

Ethernet license for channel 2. Configure your BOSCH logger to send telemetry data to IP 10.10.0.25 port 10002 or your COSWORTH logger to send telemetry data to IP 172.16.46.200 port 10002

Save

7. A message will appear, saying "Configuration saved and applied. Please POWER CYCLE the device." Press "Close". You will be redirected to the "Status" page.



- Now, to save the changes, you must reboot the device; a yellow banner on the top of the page will be present until you perform the reboot.

The screenshot shows the PowerOnTS15 web interface. At the top, there is a navigation bar with links: Status, Telemetry, Diagnostic, Reboot, and Log out. A 'REFRESHING' button is also present. A yellow banner at the top of the main content area reads 'Reboot Needed' with a 'Dismiss' button. Below the banner, the 'Status' section is visible, containing a 'System' table with the following data:

Hostname	PowerOnTS15
Model	Gyr single modem
Firmware Version	0.999
Additional channels	2
Additional modules	ETHERNET
Local Time	2026-04-09 06:45:38
Uptime	0h 1m 54s

Below the system table is the 'Telemetry status' section with the following data:

Modem 1	CONNECTED: SIGNAL WEAK
Data from RS232 channel	NONE
Connection from ethernet channel 1 (10001)	ESTABLISHED
Connection from ethernet channel 2 (10002)	LISTENING, NO DATA

The 'Network' section shows details for 'Modem1':

```

Protocol: DHCP client
Address: 192.168.225.24/24
Gateway: 192.168.225.1
DNS 1: 192.168.225.1
Expires: 11h 58m 29s
Connected: 0h 1m 31s
MAC address: 56:E4:12:7C:58:4A
  
```

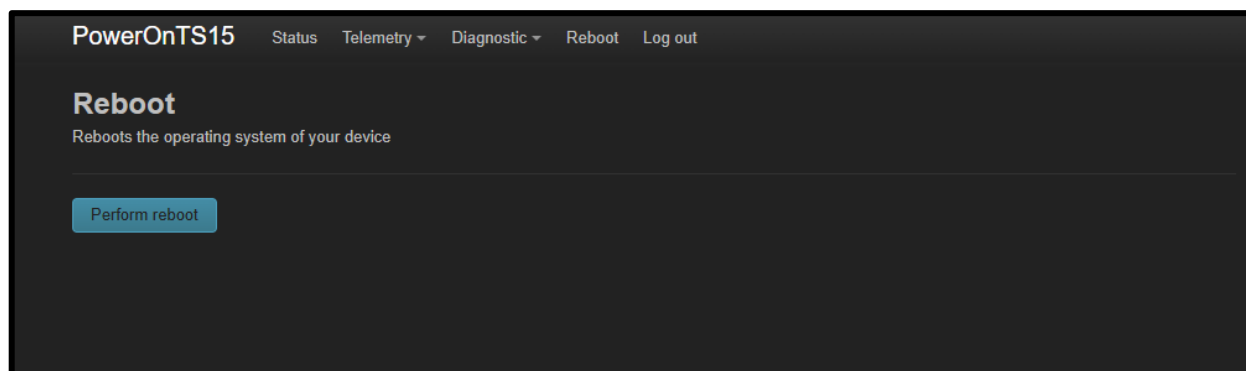
At the bottom, the 'Active Connections' section shows a progress bar for '72 / 31744 (0%)'.

One way to do it is to perform a "Power cycle". After you have done it, you must wait the following window, and then press "Log in". You will be redirected to the log in page, so you can sign in again in the Configuration page.

The screenshot shows the PowerOnTS15 web interface with a 'Session expired' dialog box overlaid. The dialog box contains the text: 'A new login is required since the authentication session expired.' and a 'Log in...' button. The background interface shows the 'Status' section with the 'System' table:

Hostname	PowerOnTS15
Model	Gyr single modem
Firmware Version	0.998
Additional channels	2

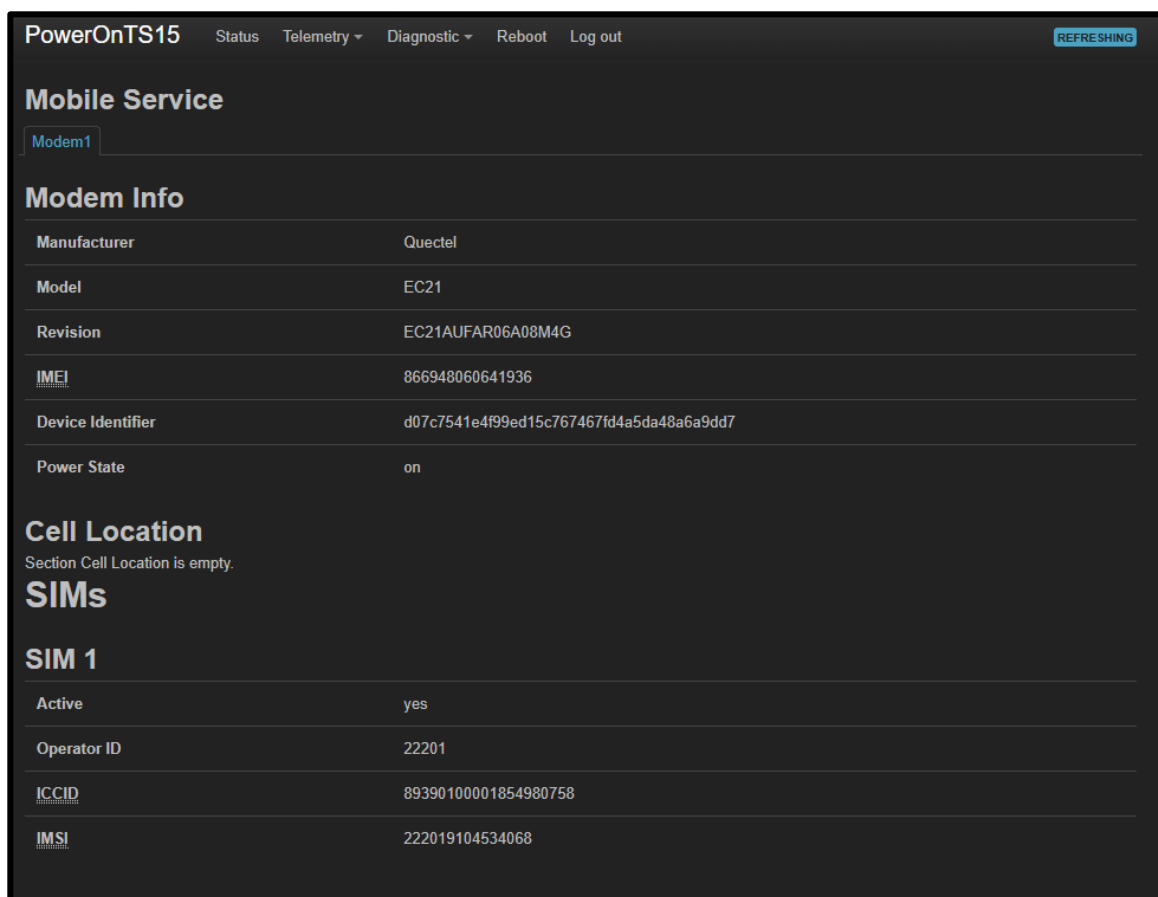
Otherwise, you can press the "Reboot" button on the top of the window, and then press "Perform reboot".



After about a minute you will be redirected to the log in page, so you can sign in again.

8.3 Check Internet connection

In the "Telemetry/Mobile Service" tab, you can check the status of your telemetry Internet connection. If everything is configured correctly, the Network Registration section will Display the Sim information:



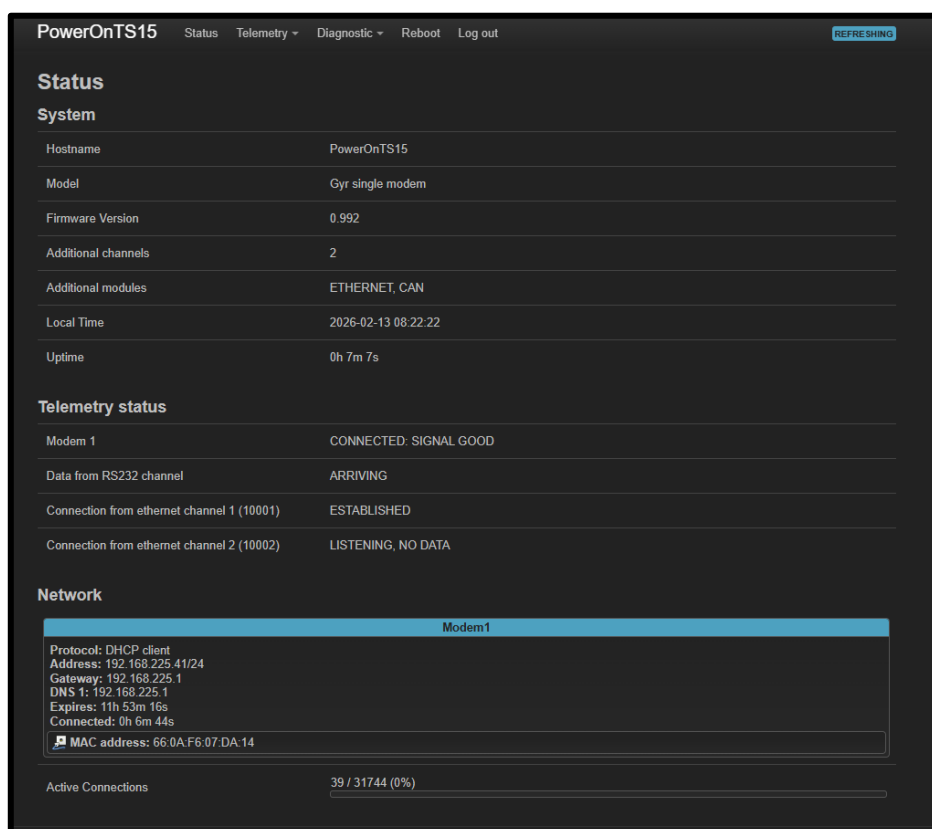
If the Sim card is missing or if it cannot be read by the telemetry, the following information will be show on the same page, at the bottom:



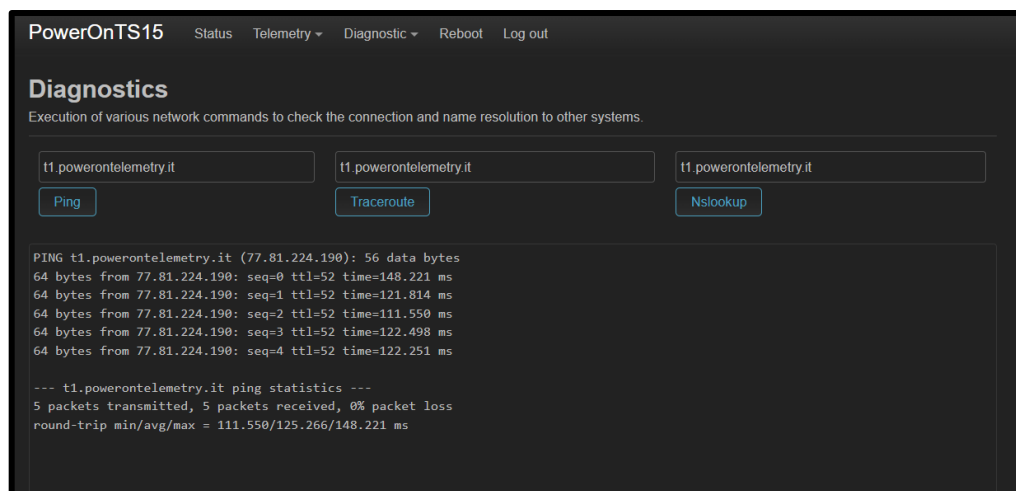
In this case, check if the Sim is inserted correctly in the telemetry. Check again if the PIN has been removed from the Sim

On the "Status" tab you can see the following information about the telemetry

- Telemetry model.
- Firmware version.
- The additional modules enabled on the telemetry.
- The connection status and signal quality.
- The data transfer status (RS232, Ethernet)
- Information about the modems.



To check the connection status, go to the "Network/Diagnostics" menu and click on "Ping". If the connection is active, the ping test with 5 packets will be successful:



9 LICENSES AND DATA TRANSMISSION

With the purchase of a TS154G or TS155G telemetry device, a basic data transmission plan is included.

The package consists of:

- one **main license**
- one **mirror license**

Each license **allows a single PC to run the PowerOn Bridge software** and connect an unlimited number of clients within the same network.

Additional mirror licenses can be obtained by upgrading the service.

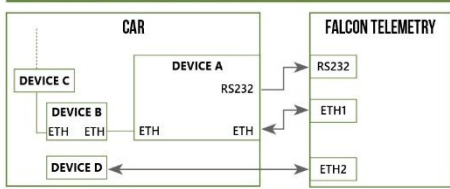
A multiple data stream service is also available as an upgrade, enabling the management of multiple data streams simultaneously.

On the next page a diagram explains the connectivity of the telemetry

CAR DATA STREAM

ETHERNET UPGRADE

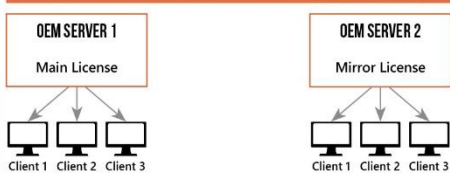
- Data streaming on RS232 and ethernet switch functionality as base setting for both telemetry models
- Data streaming on Ethernet is available by unlocking the package on Falcon GYR TS154G, while it is included by default on FALCON TS155G



EXTRA DATA STREAM

MULTIPLE STREAM DATA

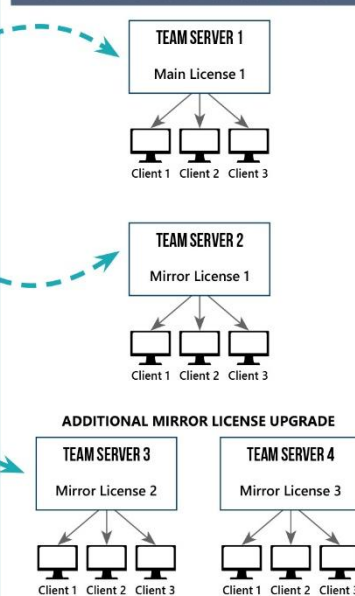
- 1 Main License + 1 Mirror License
- 2 more Mirror License (Additional Mirror License Upgrade)
- Each License allows one server PC running Power On Bridge software, with unlimited client PCs on the same network able to connect
- Multiple Stream Data needs dedicated upgrade package



MAIN DATA STREAM

BASE OPTION

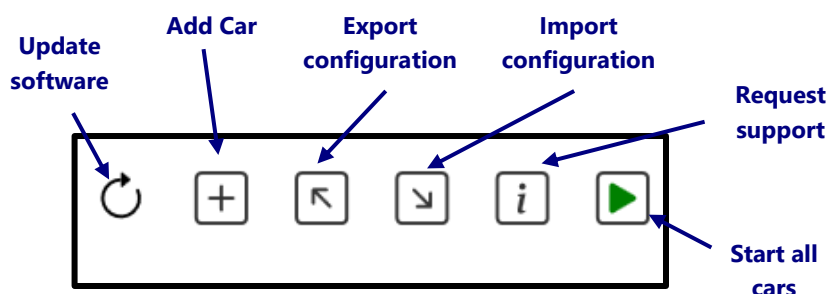
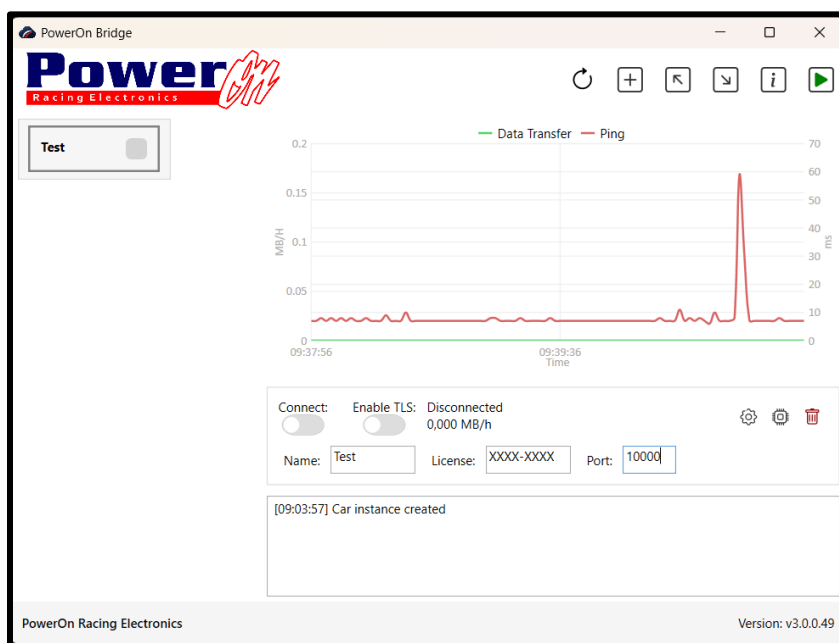
- 1 Main License + 1 Mirror License
- 2 more Mirror License (Additional Mirror License Upgrade)
- Each License allows one server PC running Power On Bridge software, with unlimited client PCs on the same network able to connect



10 POWERON BRIDGE CONFIGURATION

Install the latest available version of PowerOn Bridge (v3.1.0.0 or higher), downloadable from **t1.powerontelemetry.it**. Create a new vehicle profile, pressing the "+" on the top left of the program interface, associating the corresponding license, and set the data transmission port with the WD Server by entering:

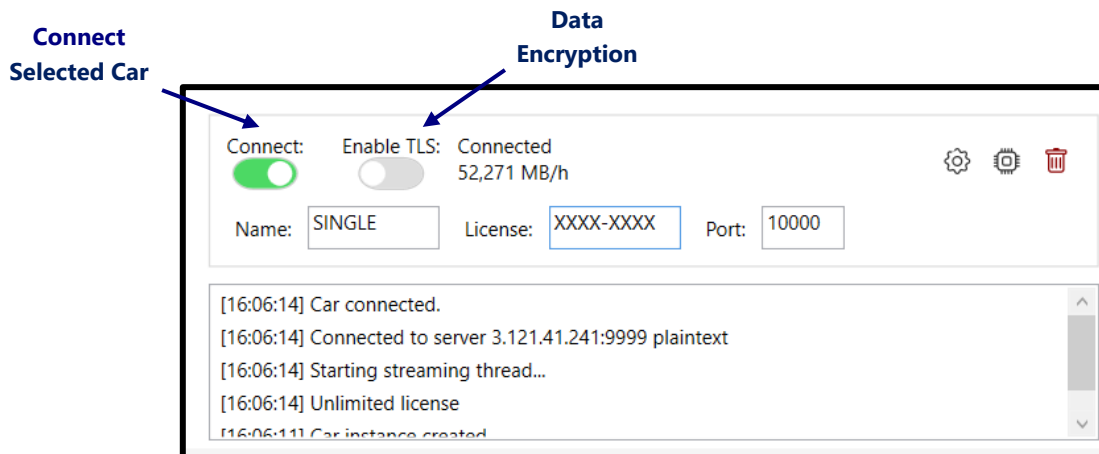
- **Name:** car identification name.
- **License:** License code that allows connection to the PowerOn Telemetry server. Enter the license code provided by Power On.
- **Port:** the port where you want to transmit the data, which can have a value from 10000 to 19999; furthermore, it must have a different value from one vehicle to another.



Once the vehicle profile has been created, it is possible to export the configuration, import it, add a new vehicle and enable/disable the reception of all created vehicles via the "Connect" button.

By clicking the "i" button, you can request support for any telemetry or PowerOn Bridge malfunctions. After completing the request form, an e-mail will be sent with to Power On, with all the information needed to resolve the issue.

The "Circular arrow" icon is used for the program update, if available.



The "Connect" switch enables/disables data reception from the selected vehicle.

The "Enable TLS" button enables/disables data encryption (**under development, connection may be unstable**).

Next to the "Enable TLS" command there is information regarding the data transfer speed (in MB/h) and the connection status.

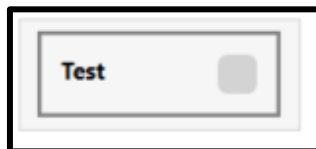
Under the "Name", "License" and "UDP Port" fields, the Log window is present, where is possible to see all the telemetry and POB messages and warnings.

Above the UDP port box, there are three buttons:

- -> Opens the Telemetry configuration page, where you can make changes to the telemetry module. (**Note: You must be connected to the telemetry module via Ethernet cable**).
- -> Allows you to update the telemetry firmware, if available. (**NB: you must be connected to the telemetry via Ethernet cable**).
- -> Remove the selected car.

The Logs window the log of events that occurred during the current PowerOn Bridge session; messages may appear in four distinct colours:

- **Black** -> system messages, remaining license validity days
- **Orange** -> warning messages, (PowerOn Bridge update present, no data received from the car, car already connected, check license code, license expiring)
- **Red** -> error messages



Next to the box containing the name of the car there is a square box, which can take on different colours, indicating different connection states:

- **Grey** -> Connection not active, connection in progress.
- **Green** -> Car connected.
- **Orange** -> No data from car.
- **Violet** -> Car is connected but no data stream is available.
- **Red** -> Impossible to connect/error.

For a complete errors list, check [PowerOnBridge errors list](#)

Warning: Make sure you always have only one PowerOn Bridge open on your PC.

11 TIPS

- Before inserting the SIM card into the modem, make sure that the PIN code has been disabled.
- The PowerOn server and PowerOn bridge continuously exchange control packets to check the status of the connection. If more packets are not received correctly, the PowerOn server closes the communication with PowerOn Bridge.
- If the Internet connection within the box is unstable, the PowerOn Bridge program may lose connection to the server. In these cases, click the "Connect" button to reconnect to the server.
- Make sure you have a stable Internet connection inside the box.

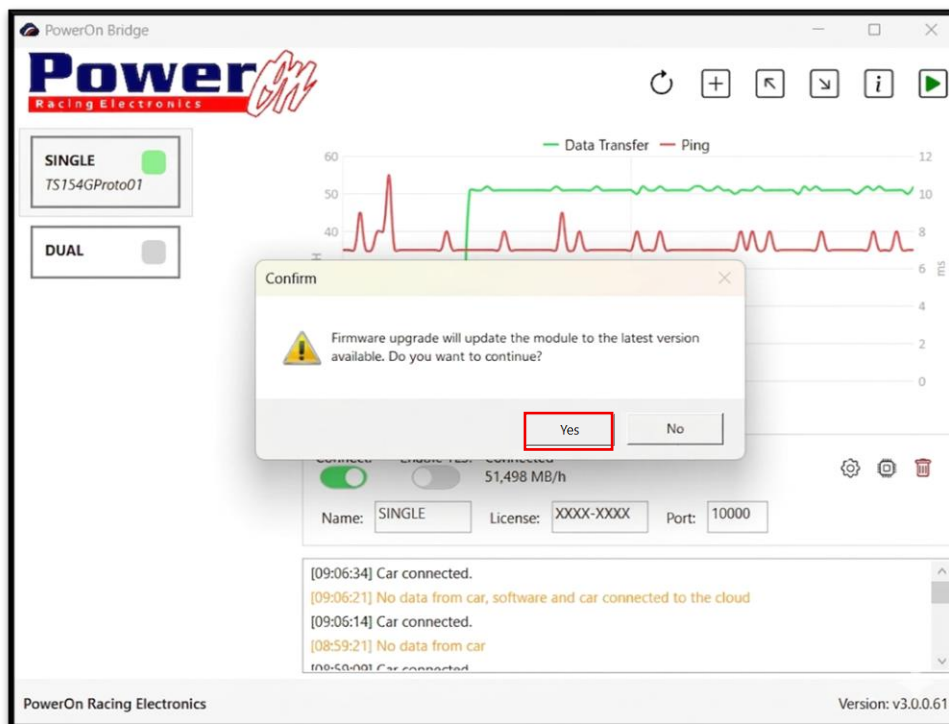
12 TELEMETRY FIRMWARE UPDATING PROCEDURE

The following part of the manual explains how to use the firmware updater for the TS154G e TS155G telemetry through the PowerOn Bridge software, allowing you to upgrade to the latest available version.

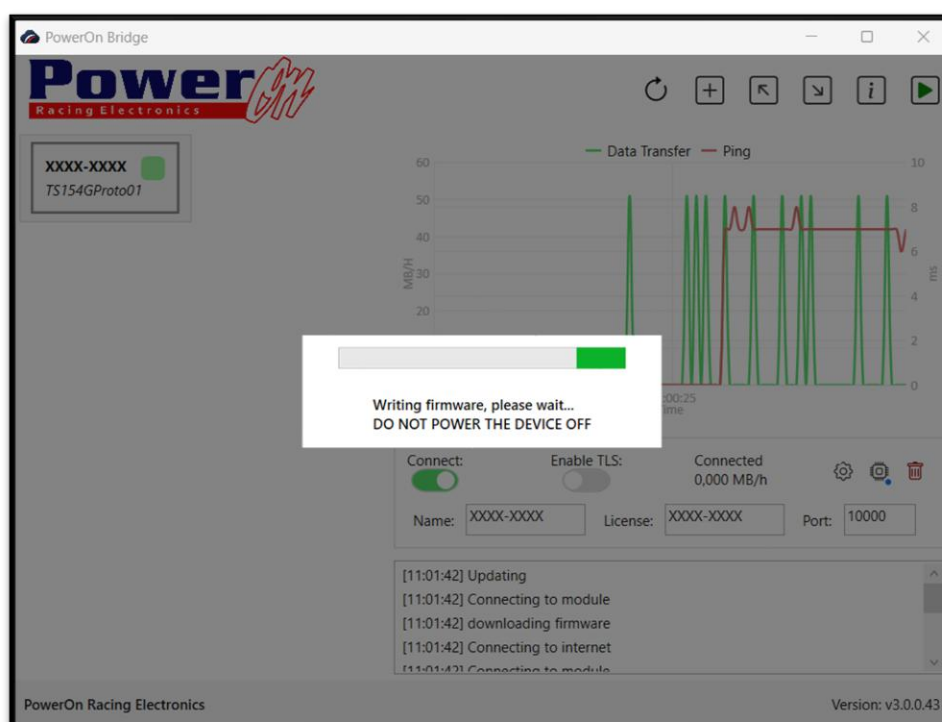
1. Open the PowerOn Bridge software, making sure telemetry is turned on and connected to your computer via **ethernet cable**, and verifying that the license is correct; the "Connect" button must be on too.
2. If a firmware update is available, a blue dot will be present on the chip icon; click on it to start the updating procedure.



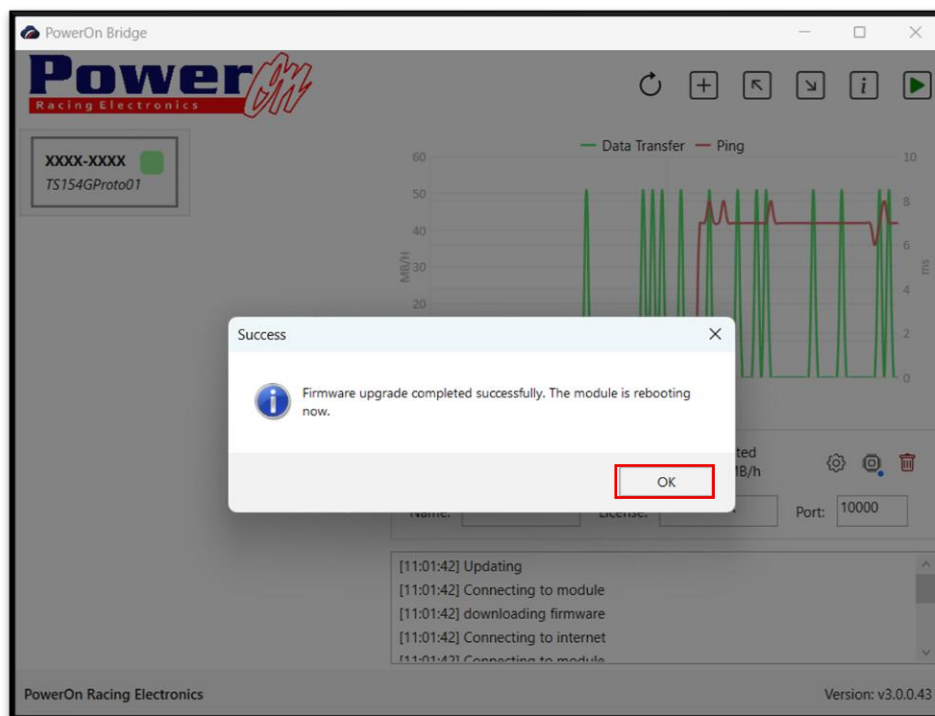
- A confirmation message will appear to start the updating procedure; press the "Yes" button.



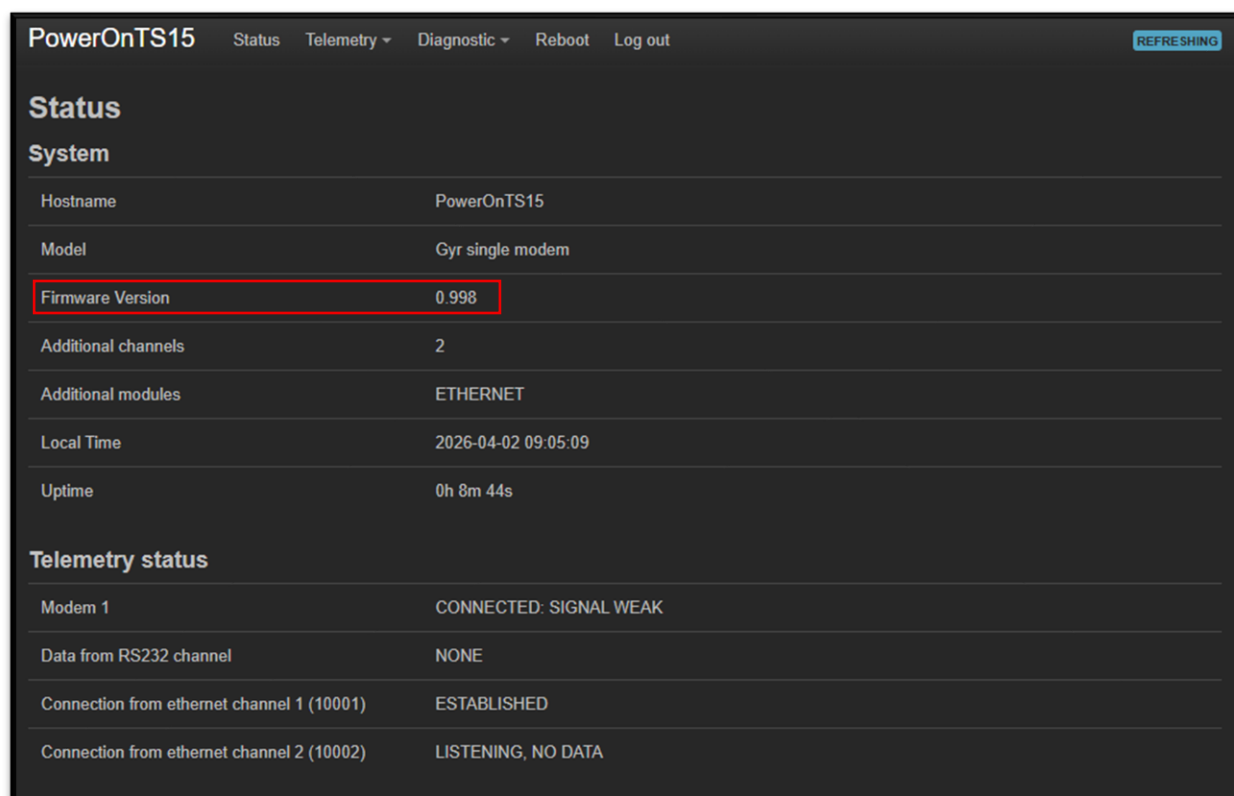
- Wait for the procedure to complete, **without disconnecting the telemetry power supply.**



- At the end of the process, a message will appear, confirming that the firmware update was successful. Press "OK" and wait for the telemetry reboot.

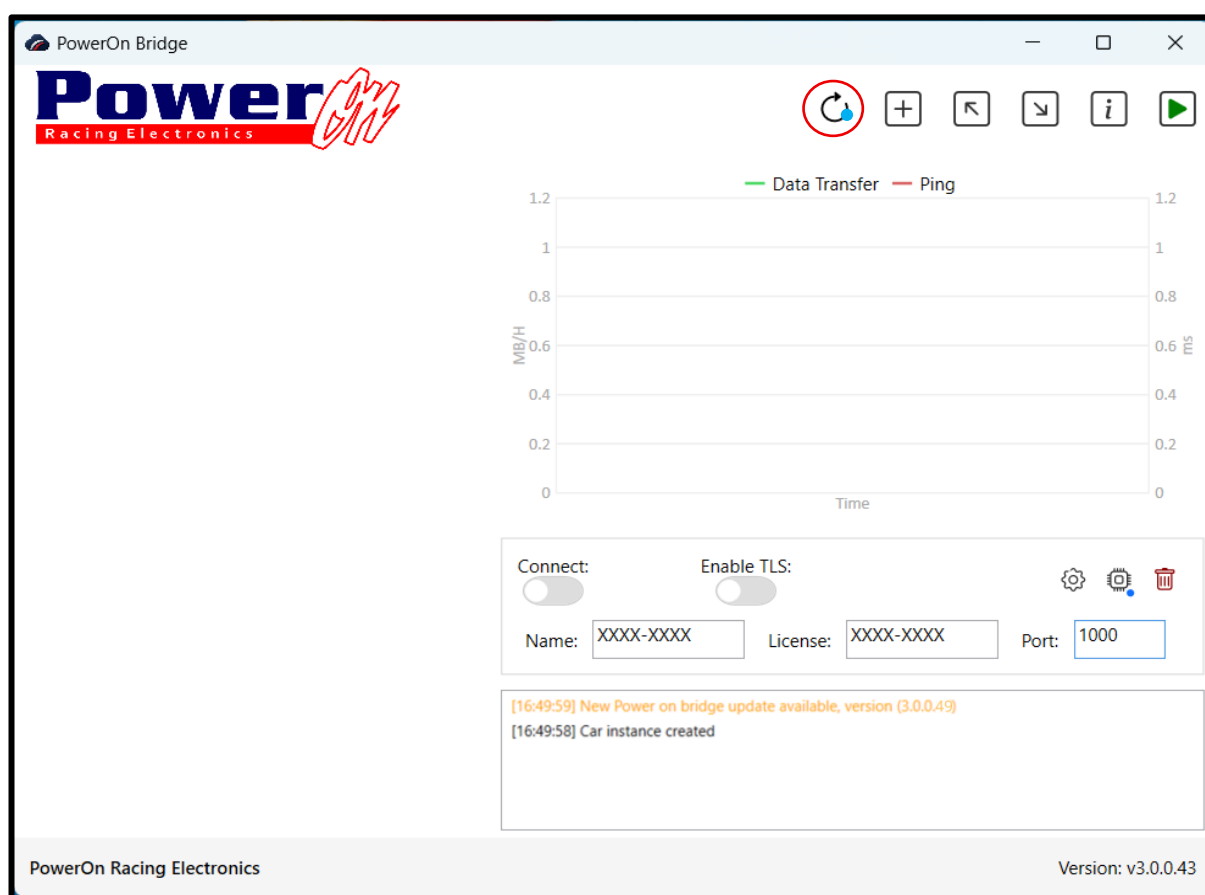


On the Configuration page, at the IP address "**192.168.1.100**", it is possible to verify the firmware version installed on the telemetry.

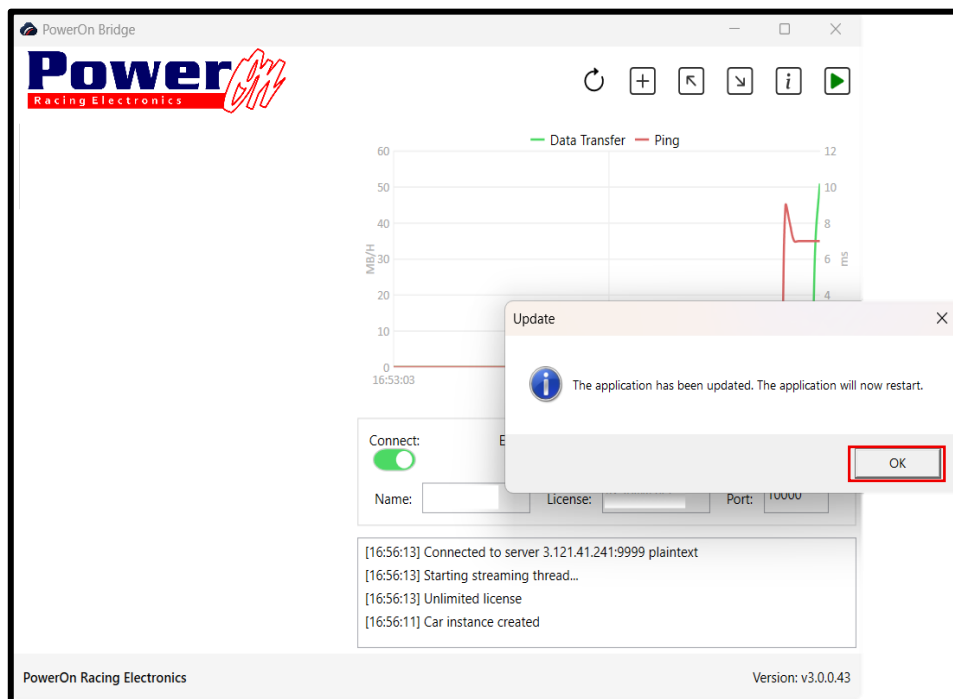


13 POWERON BRIDGE UPDATING PROCEDURE

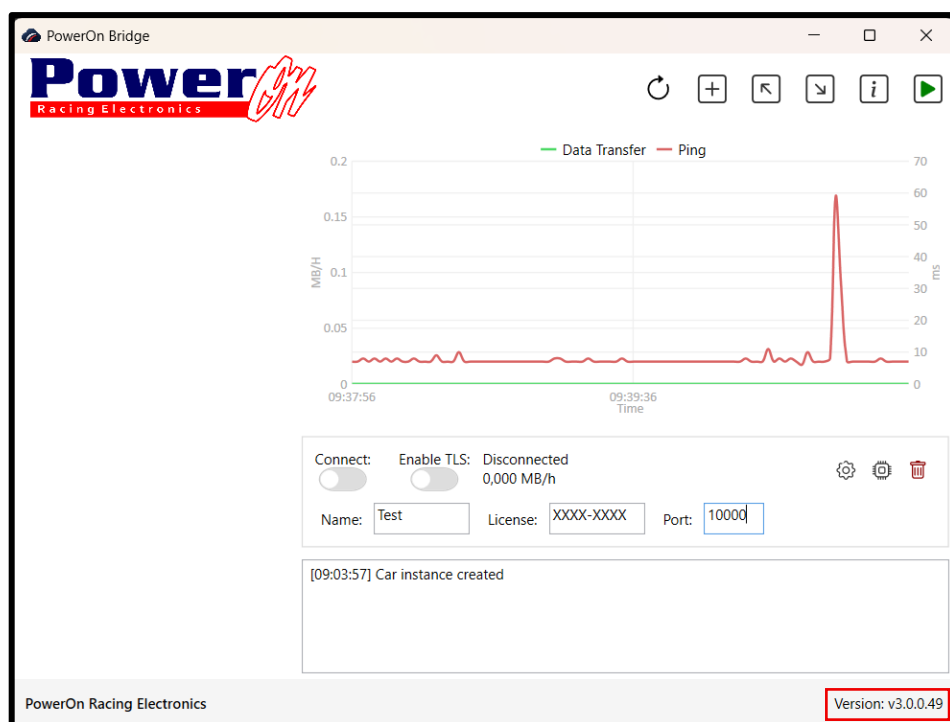
1. When opening the PowerOnBridge, in the Logs window at the bottom right, you will see a notification message saying, "**New Power on bridge update available, version (X.X.X)**". Furthermore, on the top of the window, there is an icon with a circular arrow; if an update is available, a blue dot will be present near the icon. **Press this icon to start the updating procedure and wait for the download to complete.** If the application is already up to date, an info message saying "**The application is already up to date**" will appear.



- After the updating procedure, press "OK" on the next window.



- The program will close, then you can open it again; if the update was successful, the message in the log box will have disappeared; furthermore, the serial number of the version in use at the bottom right will be updated to the latest version.



14 POWERON BRIDGE ERRORS LIST AND STATUS MESSAGES

MESSAGE	DESCRIPTION	POSSIBLE SOLUTION
Streaming already in progress or auto connect enabled...	The connection to the car is already active	--
Cannot enable car: check license code	Unable to connect to the car due to a license error	Check PowerOn license
Car is already connected or connecting.	Car is already connected or is connecting	Check if the connection to the car is already active or in the process of being activated
License check failed: ... (error detail)	Unable to verify license	Verify that the license is correct and that the network connection is on
Exception checking license: ... (error detail)	Unable to verify license	Check your network and antivirus settings
Unexpected server response: ... (error detail)	Server connection error	Check your network settings and Internet connection
License rejected by server: ... (error detail)	License rejected by server	Check if the license is correct
No data from car	No data are transmitted to the server	Check telemetry Internet connectivity, antennas connection or telemetry power state
Car is connected, but no data stream is available	No data stream is available	Check data connection between car and telemetry module
Connection timeout: no data received from server.	Connection to telemetry server expired	Check your network settings and Internet connection
Connection to the cloud is unstable. Check Internet connection.	Connection is unstable	Check computer connection status, see Ping chart
Exception communicating with server: ... (error detail)	Unable to communicate with the server	Check your network settings and Internet connection
Streaming ended, auto-reconnect in X sec	Connection ended, attempting to reconnect in X sec.	--
Auto-reconnect timer elapsed, restarting streaming...	Auto-reconnect timer expired, please re-enable data streaming	Manually reactivate the connection to the car
Error sending e-mail: ... (error detail)	Error sending support request e-mail	Try to repeat the support request submission process
New Power on bridge update available, version ...	New version of PowerOnBridge available	Download the new version of POB

MESSAGE	DESCRIPTION	POSSIBLE SOLUTION
The application has been updated. The application will now restart.	The application has been updated, please restart it	After updating the POB, restart the application to complete the update
The module is not responding, please connect your PC to the module via configuration port	Telemetry is not responding	Connect the telemetry via Ethernet to your PC to check its status
Error during firmware upgrade: ... (error detail)	Error while updating firmware	Repeat the update procedure
Cannot open the browser.	Unable to open browser	Check Internet connection
Error saving cars: ... (error detail)	Error saving selected car configuration	Try to save again the configuration in another folder
No cars found in the imported configuration.	No cars found in the configuration import file	Try to import another saved configuration
Error loading cars: ... (error detail)	Error loading car	Try loading your saved data again
Error preparing help: ... (error detail)	Error in the help request phase	Repeat the help request procedure
Select a car to remove!	No cars present for deletion	Select a car to delete

15 FIRMWARE VERSIONS

VERSION	CHANGELOG
1.0	First release

16 POWERON BRIDGE VERSIONS

VERSION	CHANGELOG
3.1.0.0	First release



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www.poweron.shop

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seguici sui social @poweronraceingelectronics

Vers.	Date	Changelog
01	14/04/2026	First release
02	22/04/2026	Added Bosch and Cosworth IP address sending data to telemetry